

Sentiment Classification of Tweets: A Comparison of Crowdsourced and Gold Annotations

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Abstract

We present a sentiment classification system for tweets that distinguishes positive, neutral, and negative opinions. Our system is built using TF-IDF feature extraction with logistic regression and is trained and evaluated on the two annotated datasets. The first one with crowdsourced labels and the other one with expert labels. Our best model, trained on the gold data, achieves 73.1% accuracy on the test set. We also analyse other factors, such as inter-annotator agreement between the two annotation sets (how consistently different people assign the same labels to the same data). To measure this we used Cohen's k , and we obtained a value of 0.446. We also identify systematic label quality issues in the crowdsourced data, and investigate the performance gap that results from training on noisy versus expert labels.

1 Introduction

During the annotation task, we can understand why some errors occurred. It was a repetitive process involving reading many tweets and assigning numerous labels. It is easy to make a typo or misclassify a tweet under these conditions. One way to improve this would be to require annotators to select from the three predefined options instead of allowing free-text input. It could also be beneficial to reduce the number of tweets each user labels, as 100 is quite a lot and fatigue may lead to additional errors.

Determining whether a piece of text expresses a positive, negative, or neutral opinion is a fundamental natural language processing task with broad applications in social media monitoring, product review analysis, and customer feedback

aggregation. In this assignment we develop a three-class sentiment classifier for Twitter data, following the requirements of the fictional company Birch. A key concern in any supervised learning pipeline is the quality of training labels. Crowdsourcing annotation at scale is quite cost-effective but often introduces inconsistencies. We therefore study how training on crowdsourced versus expert gold labels affects classifier performance.

We did the following: (i) an analysis of label quality in the crowdsourced set, (ii) a measurement of inter-annotator agreement, and (iii) a comparison of two identically configured classifiers that differ only in their training data.

2 Data

2.1 Datasets

Three csv files were provided. The crowdsourced training set contains 10,676 tweets annotated by crowd workers. The gold training set contains the same 10,676 tweets but with labels assigned by a domain expert. The test set contains 5,524 distinct tweets with gold labels.

2.2 Label Issues in the Crowdsourced Set

After inspecting the crowdsourced labels, we found out 37 distinct raw values, many of which are malformed versions of the three classes. Examples include 'Positive' (capitalised), 'nuetral', 'netural', 'negayive', 'postive'. These are caused by the crowd typos. To try to solve this we normalised all variants to lowercase and applied a deterministic mapping to the three target classes. Any label that could not be resolved would be removed. After the mapping, none of the labels could not be resolved so we preserve the full usable training set. A way of improving the data could be to make it compulsory to only choose between the labels, instead of letting the user write

its answer.

2.3 Label Distribution

We can see in Table 1 that all three datasets are class-imbalanced, with neutral being the most common class (47-49%), followed by positive (30-34%) and negative (15-22%). Notably, the gold annotations assign substantially fewer negative labels than the crowdsourced set (15.5% vs 22.2%). This suggests that crowd workers tend to over-predict negativity. Figure 1 visualises these distributions alongside the inter-annotator agreement matrix.

2.4 Inter-Annotator Agreement

Treating the crowdsourced set as one annotator and the gold set as another, we compute Cohen’s kappa. Raw label agreement is 65.5% and $k = 0.446$. Values between 0.4 and 0.6 indicate moderate agreement. This moderate score reflects ambiguity in sentiment. Tweets that could be read as neutral or positive are a common source of disagreement. The off-diagonal cells in the confusion matrix (Figure 1, centre) show that the largest disagreements occur between the neutral and positive classes, which is consistent with the subjectivity of positive statements on social media. The moderate agreement suggests that the crowdsourced labels contain a meaningful noise level, which we expect to reduce the performance of a classifier trained on them. This hypothesis is tested in Section 4.

3 Method

3.1 Text Representation

We represent each tweet as a TF-IDF vector using the scikit-learn `TfidfVectorizer` function with the following settings: First, we use unigrams and bigrams (`ngram_range=(1,2)`), to capture local context such as negation (‘not happy’). Secondly, we use sublinear TF scaling ($\log(1+tf)$), that reduces the effect of terms that appear very frequently. Thirdly, a minimum document frequency of 2, to discard terms that only appear one time and are unlikely to generalise. Finally, a vocabulary ceiling of 50,000 features. Tokens are extracted with a simple word-level pattern.

3.2 Classifier

We use multinomial logistic regression as the classifier for multiple reasons. LR classifiers handle multi-class problems via the softmax function

and produce calibrated probability estimates that are very useful in our case. Moreover, its coefficient matrix let us inspect which n-grams are most predictive of each class. An other advantage of this method is that it is computationally efficient for high-dimensional sparse inputs. We applied regularisation strength set to $C=1.0$, that was found to perform well in 5-fold cross-validation without further tuning. We used `lbfgs` as is the best optimization technique in multiclass.

Regarding the techniques, we also considered other methods like LinearSVC and Naive Bayes. LinearSVC achieved slightly higher accuracy on text tasks but does not produce probabilistic outputs. Naive Bayes is fast but its independence assumption fits bad when bigrams encode correlated context. LR offered the best balance of performance and transparency for Birch’s project.

3.3 Baseline

As a trivial baseline we used a majority-class classifier that always predicts ‘neutral’. This achieves 47.0% accuracy on the test set. Any model below this threshold would be useless in practice.

3.4 Evaluation

For evaluation we used two methods: First, 5-fold stratified crossvalidation on the training data to estimate generalisation. For the final evaluation on the held-out test set, we used per-class precision, recall, and F1-score as well as macro accuracy. We also inspect the confusion matrix and the top features per class to make sure everything works correctly.

4 Results

We can see in Table 2 that both trained models far outperform the baseline (47.0%). The model trained on gold labels achieves 73.1% test accuracy, compared to 63.2% for the crowdsourced model. The cross-validation scores are consistent with the test results, indicating that neither model is overfitting.

4.1 Confusion Matrix Analysis

We can observe in Figure 2 that it is clear that both models struggle most with the neutral class, which is expected given its broad definition. The gold model shows markedly better recall for negative tweets (78% vs 52% for the crowdsourced

Class	Crowdsourced	Gold (train)	Gold (test)
Neutral	5,079 (47.6%)	5,364 (50.2%)	2,597 (47.0%)
Positive	3,219 (30.2%)	3,652 (34.2%)	1,850 (33.5%)
Negative	2,378 (22.3%)	1,660 (15.5%)	1,077 (19.5%)

Table 1: Label distribution across the three datasets

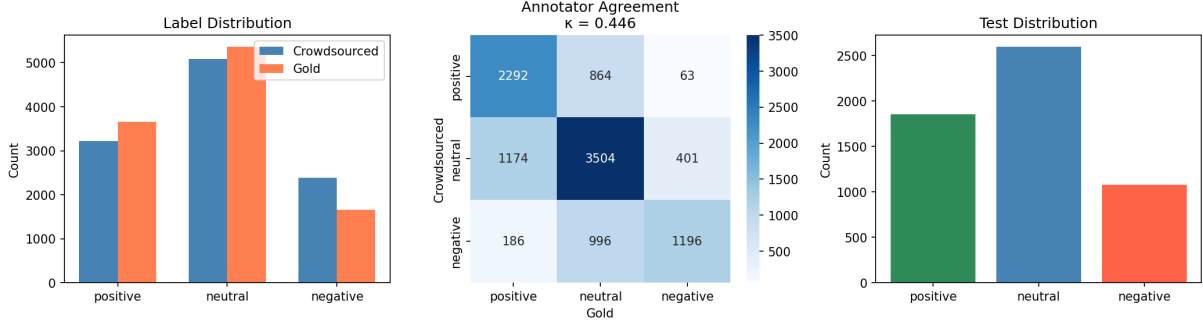


Figure 1: Left – label distribution comparison. Centre – annotator confusion matrix. Right – test set distribution.

model), likely because the gold labels cleanly separate negative from neutral sentiment, and crowdsourced annotators frequently mix the two. Both models occasionally confuse positive with neutral which, as we said, reflects the ambiguity of mildly positive language on Twitter.

4.2 Feature Analysis

In figure 3, the feature weights are reasonable: negative sentiment is associated with terms like *ira* or *worst*, although terms like *muslim* also appear. In the case of positive terms we observe *love*, *great* and other celebratory language. Finally, in neutral we get common terms, connectors and mentions.

5 Discussion

The gap in performance between the crowdsourced and gold labels makes sense, when annotators disagree a lot (which they did, $k = 0.446$), the labels are noisy. This noise hurts the model in the wrong way because it does not help it generalise better, it just teaches it the wrong patterns. You can see this most clearly in how the model handles negative tweets, it often misclassifies them as neutral, which mirrors exactly the confusion that human annotators had.

There are a few ways performance could be improved. Better text preprocessing for Twitter-specific content like hashtags, mentions and emojis would help. Tuning hyperparameters like vocabulary size and the regularisation constant could

give small gains. The biggest improvement would likely be replacing the bag-of-words features with a pre-trained model like BERTweet, though this would be slower to train and harder to interpret. None of these were implemented due to time constraints.

6 Conclusion

We developed a tweet sentiment classifier using TF-IDF features and logistic regression. Training on expert gold labels yields 73.1% test accuracy, representing a 26 percentage point improvement over the majority-class baseline. Training on crowdsourced labels instead reduces accuracy by 9.9 percentage points, a difference directly attributable to lower annotation quality and systematic labeling errors in the crowdsourced set. These results demonstrate that label quality has a measurable and practically significant effect on classifier performance. For any real-world sentiment analysis system, investing in high-quality annotation is not optional, it directly determines how well the model works.

Model	5-fold CV Acc	Test Accuracy
Baseline (majority class)	-	0.4701
LR trained on Crowdsourced data	0.5768 ± 0.0093	0.6320
LR trained on Gold data	0.6393 ± 0.0063	0.7310

Table 2: Performance comparison of all systems.

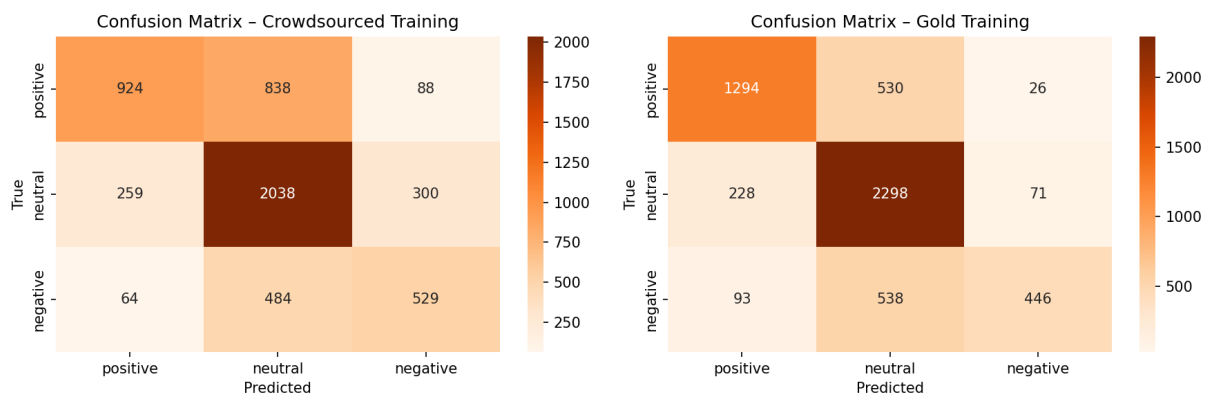


Figure 2: Confusion matrices for the crowdsourced-trained (left) and gold-trained (right) classifiers.

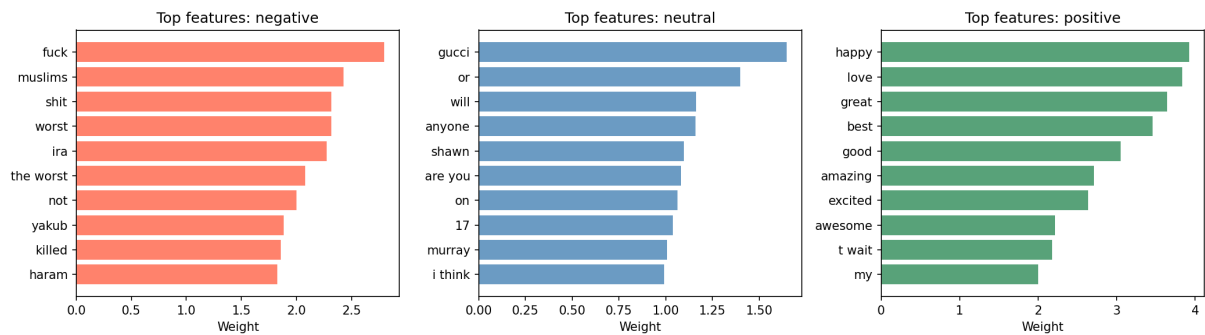


Figure 3: Top 10 TF-IDF weighted features per class (gold model).